

```
import numpy as numpy
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
df=pd.read_csv('Student Performance_dataset.csv')
```

```
df.head()
```

|   | school | sex | age | address | famsize | Pstatus | Medu | Fedu | Mjob    | Fjob     | reason | guardian | traveltime | studytime | failures | schoolsup | famsup | paid | activities | nursery | higher | internet | romantic | famrel | freetime | goout | Dalc | Walc | health | absences | G1 | G2 | G3 |  |
|---|--------|-----|-----|---------|---------|---------|------|------|---------|----------|--------|----------|------------|-----------|----------|-----------|--------|------|------------|---------|--------|----------|----------|--------|----------|-------|------|------|--------|----------|----|----|----|--|
| 0 | GP     | F   | 18  | U       | GT3     | A       | 4    | 4    | at_home | teacher  | course | mother   | 2          | 2         | 0        | yes       | no     | no   | no         | yes     | yes    | no       | no       | 4      | 3        | 4     | 1    | 1    | 3      | 6        | 5  | 6  | 6  |  |
| 1 | GP     | F   | 17  | U       | GT3     | T       | 1    | 1    | at_home | other    | course | father   | 1          | 2         | 0        | no        | yes    | no   | no         | no      | yes    | yes      | no       | 5      | 3        | 3     | 1    | 1    | 3      | 4        | 5  | 5  | 6  |  |
| 2 | GP     | F   | 15  | U       | LE3     | T       | 1    | 1    | at_home | other    | other  | mother   | 1          | 2         | 3        | yes       | no     | yes  | no         | yes     | yes    | yes      | no       | 4      | 3        | 2     | 2    | 3    | 3      | 10       | 7  | 8  | 10 |  |
| 3 | GP     | F   | 15  | U       | GT3     | T       | 4    | 2    | health  | services | home   | mother   | 1          | 3         | 0        | no        | yes    | yes  | yes        | yes     | yes    | yes      | yes      | 3      | 2        | 2     | 1    | 1    | 5      | 2        | 15 | 14 | 15 |  |
| 4 | GP     | F   | 16  | U       | GT3     | T       | 3    | 3    | other   | other    | home   | father   | 1          | 2         | 0        | no        | yes    | yes  | no         | yes     | yes    | no       | no       | 4      | 3        | 2     | 1    | 2    | 5      | 4        | 6  | 10 | 10 |  |

```
df.tail()
```

|     | school | sex | age | address | famsize | Pstatus | Medu | Fedu | Mjob     | Fjob     | reason | guardian | traveltime | studytime | failures | schoolsup | famsup | paid | activities | nursery | higher | internet | romantic | famrel | freetime | goout | Dalc | Walc | health | absences | G1 | G2 | G3 |  |
|-----|--------|-----|-----|---------|---------|---------|------|------|----------|----------|--------|----------|------------|-----------|----------|-----------|--------|------|------------|---------|--------|----------|----------|--------|----------|-------|------|------|--------|----------|----|----|----|--|
| 390 | MS     | M   | 20  | U       | LE3     | A       | 2    | 2    | services | services | course | other    | 1          | 2         | 2        | no        | yes    | yes  | no         | yes     | yes    | no       | no       | 5      | 5        | 4     | 4    | 5    | 4      | 11       | 9  | 9  | 9  |  |
| 391 | MS     | M   | 17  | U       | LE3     | T       | 3    | 1    | services | services | course | mother   | 2          | 1         | 0        | no        | no     | no   | no         | no      | yes    | yes      | no       | 2      | 4        | 5     | 3    | 4    | 2      | 3        | 14 | 16 | 16 |  |
| 392 | MS     | M   | 21  | R       | GT3     | T       | 1    | 1    | other    | other    | course | other    | 1          | 1         | 3        | no        | no     | no   | no         | no      | yes    | no       | no       | 5      | 5        | 3     | 3    | 3    | 3      | 3        | 10 | 8  | 7  |  |
| 393 | MS     | M   | 18  | R       | LE3     | T       | 3    | 2    | services | other    | course | mother   | 3          | 1         | 0        | no        | no     | no   | no         | no      | yes    | yes      | no       | 4      | 4        | 1     | 3    | 4    | 5      | 0        | 11 | 12 | 10 |  |
| 394 | MS     | M   | 19  | U       | LE3     | T       | 1    | 1    | other    | at_home  | course | father   | 1          | 1         | 0        | no        | no     | no   | no         | yes     | yes    | yes      | no       | 3      | 2        | 3     | 3    | 3    | 5      | 5        | 8  | 9  | 9  |  |

```
df.shape
```

```
(395, 33)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column      Non-Null Count  Dtype
---  -
0   school      395 non-null    object
1   sex         395 non-null    object
2   age         395 non-null    int64
3   address     395 non-null    object
4   famsize     395 non-null    object
5   Pstatus     395 non-null    object
6   Medu       395 non-null    int64
7   Fedu       395 non-null    int64
8   Mjob       395 non-null    object
9   Fjob       395 non-null    object
10  reason      395 non-null    object
11  guardian    395 non-null    object
12  traveltime  395 non-null    int64
13  studytime   395 non-null    int64
14  failures    395 non-null    int64
15  schoolsup   395 non-null    object
16  famsup      395 non-null    object
17  paid        395 non-null    object
18  activities  395 non-null    object
19  nursery     395 non-null    object
20  higher      395 non-null    object
21  internet    395 non-null    object
22  romantic    395 non-null    object
23  famrel      395 non-null    int64
24  freetime    395 non-null    int64
25  goout       395 non-null    int64
26  Dalc        395 non-null    int64
27  Walc        395 non-null    int64
28  health      395 non-null    int64
29  absences    395 non-null    int64
30  G1          395 non-null    int64
31  G2          395 non-null    int64
32  G3          395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

df.describe()

|       | age        | Medu       | Fedu       | traveltime | studytime  | failures   | famrel     | freetime   | goout      | Dalc       | Walc       | health     | absences   | G1         | G2         | G3         |  |
|-------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------------------------------------------------------------------------------|
| count | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 |                                                                                     |
| mean  | 16.696203  | 2.749367   | 2.521519   | 1.448101   | 2.035443   | 0.334177   | 3.944304   | 3.235443   | 3.108861   | 1.481013   | 2.291139   | 3.554430   | 5.708861   | 10.908861  | 10.713924  | 10.415190  |                                                                                     |
| std   | 1.276043   | 1.094735   | 1.088201   | 0.697505   | 0.839240   | 0.743651   | 0.896659   | 0.998862   | 1.113278   | 0.890741   | 1.287897   | 1.390303   | 8.003096   | 3.319195   | 3.761505   | 4.581443   |                                                                                     |
| min   | 15.000000  | 0.000000   | 0.000000   | 1.000000   | 1.000000   | 0.000000   | 1.000000   | 1.000000   | 1.000000   | 1.000000   | 1.000000   | 1.000000   | 0.000000   | 3.000000   | 0.000000   | 0.000000   |                                                                                     |
| 25%   | 16.000000  | 2.000000   | 2.000000   | 1.000000   | 1.000000   | 0.000000   | 4.000000   | 3.000000   | 2.000000   | 1.000000   | 1.000000   | 3.000000   | 0.000000   | 8.000000   | 9.000000   | 8.000000   |                                                                                     |
| 50%   | 17.000000  | 3.000000   | 2.000000   | 1.000000   | 2.000000   | 0.000000   | 4.000000   | 3.000000   | 3.000000   | 1.000000   | 2.000000   | 4.000000   | 4.000000   | 11.000000  | 11.000000  | 11.000000  |                                                                                     |
| 75%   | 18.000000  | 4.000000   | 3.000000   | 2.000000   | 2.000000   | 0.000000   | 5.000000   | 4.000000   | 4.000000   | 2.000000   | 3.000000   | 5.000000   | 8.000000   | 13.000000  | 13.000000  | 14.000000  |                                                                                     |
| max   | 22.000000  | 4.000000   | 4.000000   | 4.000000   | 4.000000   | 3.000000   | 5.000000   | 5.000000   | 5.000000   | 5.000000   | 5.000000   | 5.000000   | 75.000000  | 19.000000  | 19.000000  | 20.000000  |                                                                                     |

df.isnull().sum()

|            | 0 |
|------------|---|
| school     | 0 |
| sex        | 0 |
| age        | 0 |
| address    | 0 |
| famsize    | 0 |
| Pstatus    | 0 |
| Medu       | 0 |
| Fedu       | 0 |
| Mjob       | 0 |
| Fjob       | 0 |
| reason     | 0 |
| guardian   | 0 |
| traveltime | 0 |
| studytime  | 0 |
| failures   | 0 |
| schoolsup  | 0 |
| famsup     | 0 |
| paid       | 0 |
| activities | 0 |
| nursery    | 0 |
| higher     | 0 |
| internet   | 0 |
| romantic   | 0 |
| famrel     | 0 |
| freetime   | 0 |
| goout      | 0 |
| Dalc       | 0 |
| Walc       | 0 |
| health     | 0 |
| absences   | 0 |
| G1         | 0 |
| G2         | 0 |
| G3         | 0 |

dtype: int64

df

|     | school | sex | age | address | famsize | Pstatus | Medu | Fedu | Mjob     | Fjob     | reason | guardian | traveltime | studytime | failures | schoolsup | famsup | paid | activities | nursery | higher | internet | romantic | famrel | freetime | goout | Dalc | Walc | health | absences | G1  | G2  | G3  |
|-----|--------|-----|-----|---------|---------|---------|------|------|----------|----------|--------|----------|------------|-----------|----------|-----------|--------|------|------------|---------|--------|----------|----------|--------|----------|-------|------|------|--------|----------|-----|-----|-----|
| 0   | GP     | F   | 18  | U       | GT3     | A       | 4    | 4    | at_home  | teacher  | course | mother   | 2          | 2         | 0        | yes       | no     | no   | no         | yes     | yes    | no       | no       | 4      | 3        | 4     | 1    | 1    | 3      | 6        | 5   | 6   | 6   |
| 1   | GP     | F   | 17  | U       | GT3     | T       | 1    | 1    | at_home  | other    | course | father   | 1          | 2         | 0        | no        | yes    | no   | no         | no      | yes    | yes      | no       | 5      | 3        | 3     | 1    | 1    | 3      | 4        | 5   | 5   | 6   |
| 2   | GP     | F   | 15  | U       | LE3     | T       | 1    | 1    | at_home  | other    | other  | mother   | 1          | 2         | 3        | yes       | no     | yes  | no         | yes     | yes    | yes      | no       | 4      | 3        | 2     | 2    | 3    | 3      | 10       | 7   | 8   | 10  |
| 3   | GP     | F   | 15  | U       | GT3     | T       | 4    | 2    | health   | services | home   | mother   | 1          | 3         | 0        | no        | yes    | yes  | yes        | yes     | yes    | yes      | yes      | 3      | 2        | 2     | 1    | 1    | 5      | 2        | 15  | 14  | 15  |
| 4   | GP     | F   | 16  | U       | GT3     | T       | 3    | 3    | other    | other    | home   | father   | 1          | 2         | 0        | no        | yes    | yes  | no         | yes     | yes    | no       | no       | 4      | 3        | 2     | 1    | 2    | 5      | 4        | 6   | 10  | 10  |
| ... | ...    | ... | ... | ...     | ...     | ...     | ...  | ...  | ...      | ...      | ...    | ...      | ...        | ...       | ...      | ...       | ...    | ...  | ...        | ...     | ...    | ...      | ...      | ...    | ...      | ...   | ...  | ...  | ...    | ...      | ... | ... | ... |
| 390 | MS     | M   | 20  | U       | LE3     | A       | 2    | 2    | services | services | course | other    | 1          | 2         | 2        | no        | yes    | yes  | no         | yes     | yes    | no       | no       | 5      | 5        | 4     | 4    | 5    | 4      | 11       | 9   | 9   | 9   |
| 391 | MS     | M   | 17  | U       | LE3     | T       | 3    | 1    | services | services | course | mother   | 2          | 1         | 0        | no        | no     | no   | no         | no      | yes    | yes      | no       | 2      | 4        | 5     | 3    | 4    | 2      | 3        | 14  | 16  | 16  |
| 392 | MS     | M   | 21  | R       | GT3     | T       | 1    | 1    | other    | other    | course | other    | 1          | 1         | 3        | no        | no     | no   | no         | no      | yes    | no       | no       | 5      | 5        | 3     | 3    | 3    | 3      | 3        | 10  | 8   | 7   |
| 393 | MS     | M   | 18  | R       | LE3     | T       | 3    | 2    | services | other    | course | mother   | 3          | 1         | 0        | no        | no     | no   | no         | no      | yes    | yes      | no       | 4      | 4        | 1     | 3    | 4    | 5      | 0        | 11  | 12  | 10  |
| 394 | MS     | M   | 19  | U       | LE3     | T       | 1    | 1    | other    | at_home  | course | father   | 1          | 1         | 0        | no        | no     | no   | no         | yes     | yes    | yes      | no       | 3      | 2        | 3     | 3    | 3    | 5      | 5        | 8   | 9   | 9   |

395 rows × 33 columns

```
df=pd.read_csv('Student Performance_dataset.csv')
x=df[['school','sex','age','address','famsize','Pstatus','Medu','Fedu','Mjob','Fjob','reason','guardian','traveltime','studytime','failures','schoolsup','famsup','paid','activities','nursery','higher','internet','romantic','famrel','freetime','goout','Dalc','Walc','health','absence:
y=df[['G1','G2','G3']]
x = pd.get_dummies(x, drop_first=True)
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
model=LinearRegression()
model.fit(x_train,y_train)
```

LinearRegression ⓘ ?

LinearRegression()

```
y_pred=model.predict(x_test)
mse=mean_squared_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(mse)
print(r2)
```

```
13.222904363078586
0.1864181949692726
```

```
new=pd.DataFrame({
    'school': ['GP'],
    'sex': ['F'],
    'age': [18],
    'address': ['U'],
    'famsize': ['GT3'],
    'Pstatus': ['A'],
    'Medu': [4],
    'Fedu': [4],
    'Mjob': ['at_home'],
    'Fjob': ['teacher'],
    'reason': ['course'],
    'guardian': ['mother'],
    'traveltime': [2],
    'studytime': [2],
    'failures': [0],
    'schoolsup': ['yes'],
    'famsup': ['no'],
    'paid': ['no'],
    'activities': ['no'],
    'nursery': ['yes'],
    'higher': ['yes'],
    'internet': ['no'],
    'romantic': ['no'],
    'famrel': [4],
    'freetime': [3],
    'goout': [4],
    'Dalc': [1],
    'Walc': [1],
    'health': [3],
    'absences': [None] # Placeholder for missing 'absences'
})
# Apply one-hot encoding to the new data, similar to how it was done for the training data
new_processed = pd.get_dummies(new, drop_first=True)
```



```
# Ensure the columns of the new_processed DataFrame match x_train
# Add missing columns with 0 and remove extra columns
missing_cols = set(x_train.columns) - set(new_processed.columns)
for c in missing_cols:
    new_processed[c] = 0

# Ensure the order of columns is the same as in x_train
new_processed = new_processed[x_train.columns]

predicted_count=model.predict(new_processed)
print("predicted_count",predicted_count[0])
```

predicted\_count [10.51656274 8.92235316 8.48476426]